Concept and Technologies of AI:

HDI-Analysis

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Course: 5CS037 – Concepts and Technologies of AI

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GitHub Repository: https://github.com/np03cs4a240277-alt/Assesment.git

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Introduction:

Human development is the process by which a person's skills and choices evolve. It enhances the likelihood that people will live long, healthy, and happy lives. The three essential components of human development at the national level that comprise the HDI composite index are knowledge, a long and healthy life, and a respectable standard of living. Consequently, the normalized indices of those three dimension cuts show up as an HDI weighted average. It merely captures a tiny portion of human development. They don't concentrate on issues like empowerment, human security, poverty, or inequality. The Human Resources Department has additional indices that would make it more representative of some of the most important problems in human development, including inequality, gender disparity, and poverty.

Objectives of Analysis:

• To explore and analyze the data for the year 2022 based on the HDI and then verify if there is any change between the years 2020 to 2022.

• The best ands wrost performing performing nations.

• Gaps and outliers between economic data and the HDI

• A regional analysis of the Middle East and South Asia.

Scope of Report:

It involves statistical analyse, data cleaning, visualizations, correlation analysis, outlier detection, and regional comparison in the given HDI datasets. Google Collab was used to generate all calculations, graphics, and intermediate results. The submitted notebook and the associated GitHub repository contain the whole analysis workflow, including code and outputs.

Problem 1A

Approach and methodology:

- HDI data for 2022 was extracted, and a cleaned subset (hdi_2022_df) was produced
- Removed aggregate regions (e.g., World, South Asia) and standardized country names Numerical columns were converted to the proper data types.
- Converted numeric columns to appropriate data types
- Rows with missing HDI values were eliminated.
- The mean, median, and standard deviation were calculated as summary statistics.
- Determined which nations had the highest and lowest HDI
- Sorting by GNI per capita, filtered nations with HDI > 0.800
- UNDP HDI thresholds were used to categorize nations, and the dataset was saved.

Key Results :

. **Final dataset:** 193 countries, 30 variables

. **HDI statistics (2022):**

Mean	Median	Standard Deviation
0.7228872549019609	0.7395	0.1530288038642782

- **Highest HDI:** Switzerland (0.967).
- **Lowest HDI:** Somalia (0.38).
- **Top Countries with Very High HDI by GNI:** Liechtenstein, Qatar, Singapore, Ireland, Luxembourg etc.
- **HDI Category Distribution:**

HDI Category	Count
Very High	69
High	49
Medium	42
Low	33

Visualizations and Tables:

Summary statistics table for HDI (2022):

```
hdi_2022_df['hdi'].mean(), hdi_2022_df['hdi'].median(), hdi_2022_df['hdi'].std()

(np.float64(0.7228872549019609), 0.7395, 0.15302880386427825)
```

Countries with highest HDI(2022)

Unnamed: 0	5610
iso3	CHE
country	Switzerland
year	2022
hdi	0.967

Countries with Lowest HDI(2022):

Unnamed: 0	5346
iso3	SOM
country	Somalia
year	2022
hdi	0.38

Interpretation and Discussion:

The 2022 HDI distribution reveals that the mostof nations lie in the High or Very High development brackets, and only a few still linger in the Low category.

Indeed, income is a major factor linked to HDI, but it is not the only one, since countries at comparable income levels may have different development results depending on their health and education sectors.

2.2 Problem 1B

Approach and methodology:

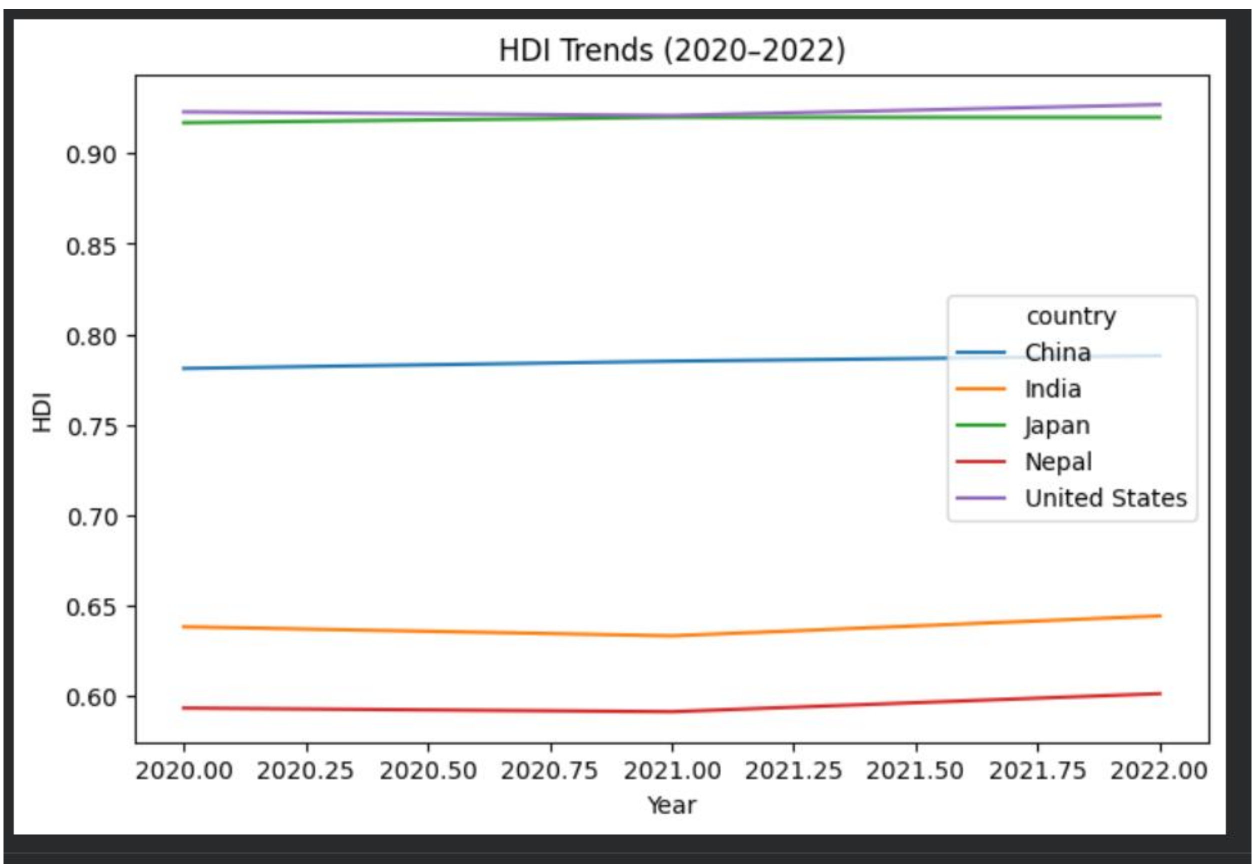
The Human Development Index (HDI) data for 2020–2022 was obtained and cleaned. Trends were evaluated using bar charts for regional averages, scatter plots examining the relationship between HDI and GNI per capita, and line plots for individual countries.

Key Results:

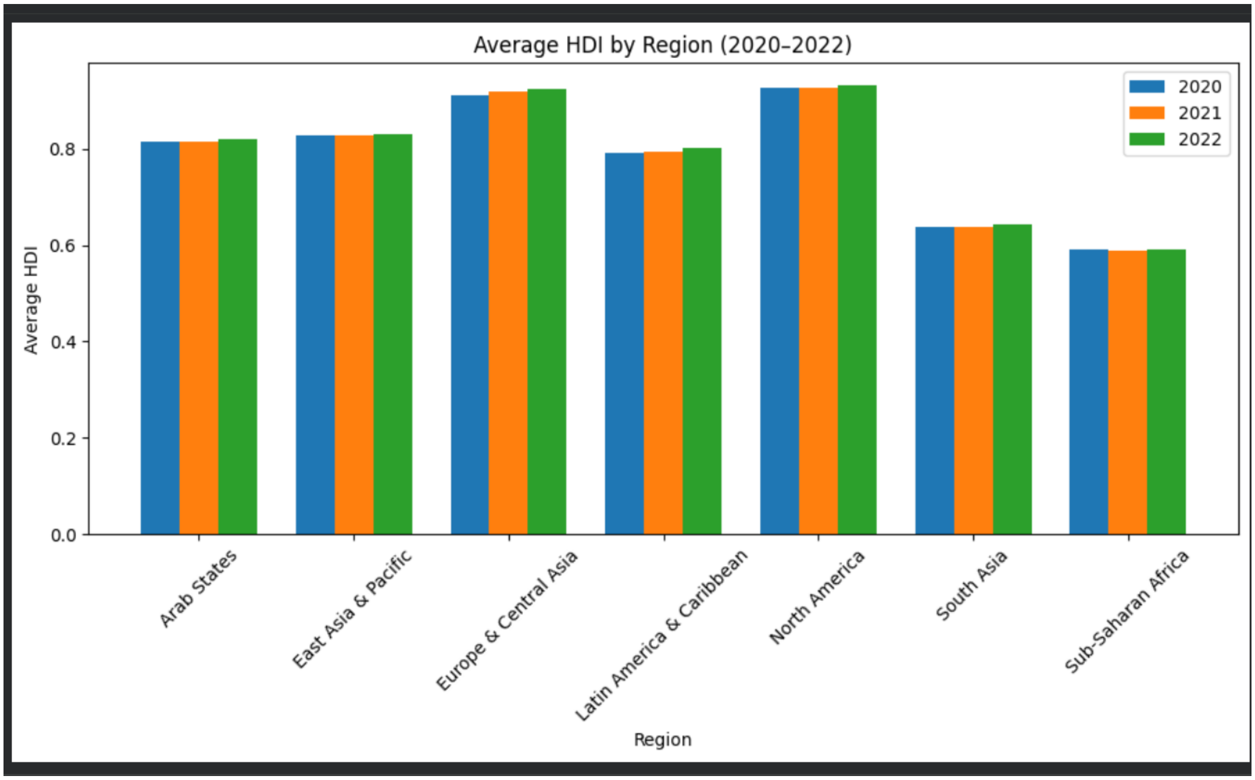
- Most likely due of the COVID-19 pandemic, the HDI decreased in several countries during 2020–2021.
- Partial recovery is observed in 2022, though progress remains uneven
- North America and Europe consistently show the highest average HDI values

Visualizations and Tables:

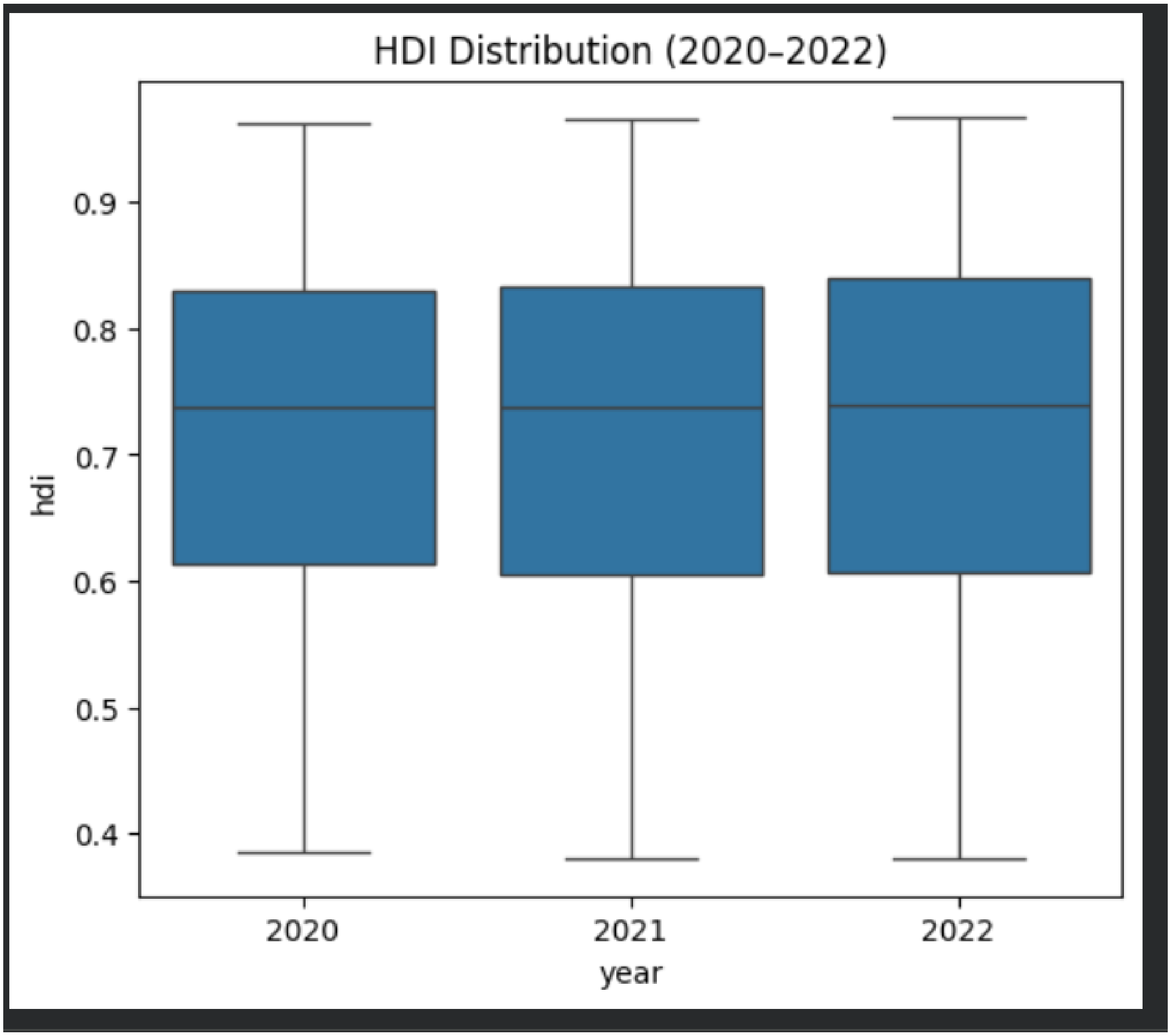
- Line chart showing HDI trends for selected countries:



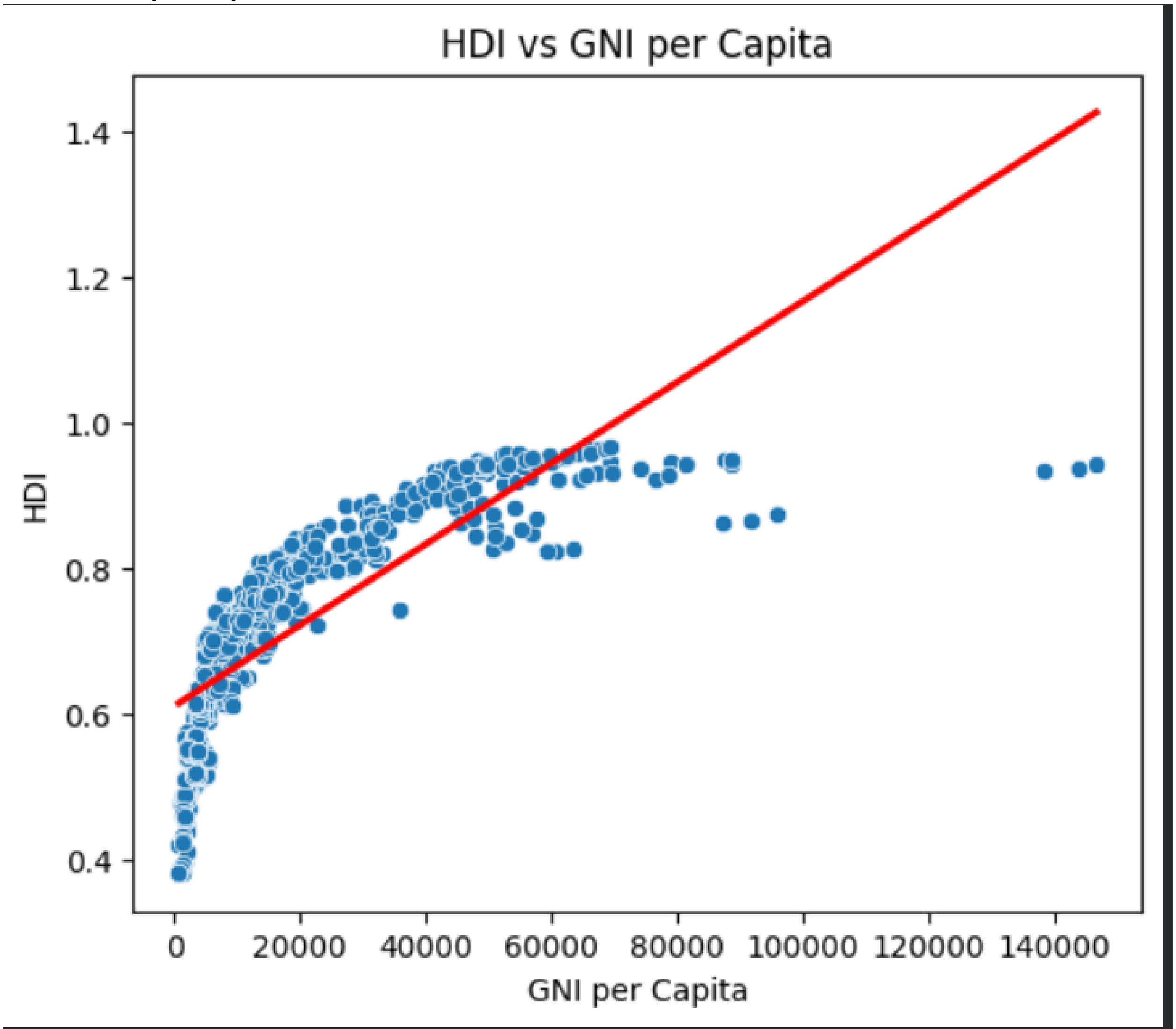
- bar chart comparing average HDI across regions for each year:



- HDI Distribution for 2020, 2021, and 2022:



- **HDI vs. GNI per Capita:**



Interpretation and Discussion:

According to the analysis, certain countries experienced setbacks during the pandemic, and worldwide HDI growth halted. Developing countries recovered more slowly than higher-income regions. At higher income levels, the connection between income and HDI exhibits diminishing returns.

2.3 Problem 2 – Advanced HDI Exploration

Approach and methodology:

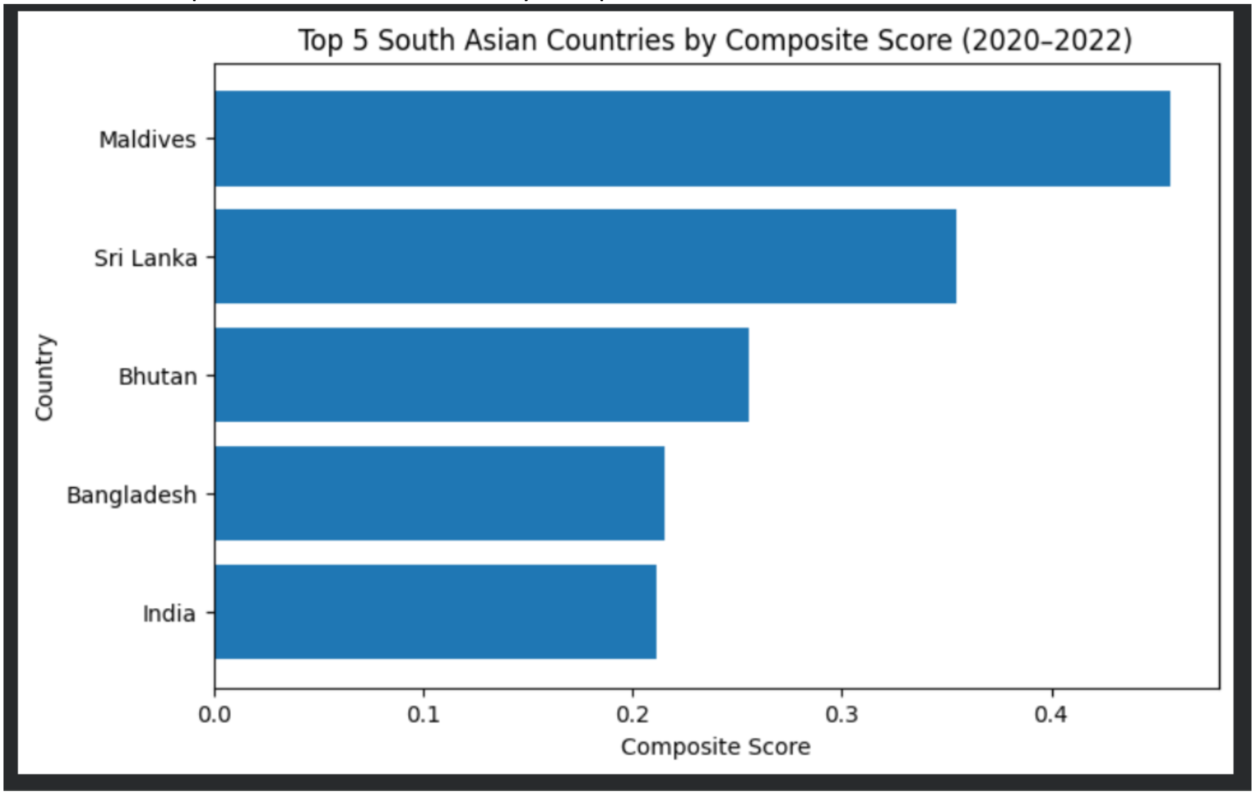
Afghanistan, Bangladesh, Bhutan, India, Maldives, Nepal, Pakistan, and Sri Lanka are all included in this section. Life expectancy and GNI per capita indicators were used to produce a composite development score. For outlier detection, correlation analysis, and gap analysis between GNI and HDI, we employed the 1.5×IQR approach.

Key Results

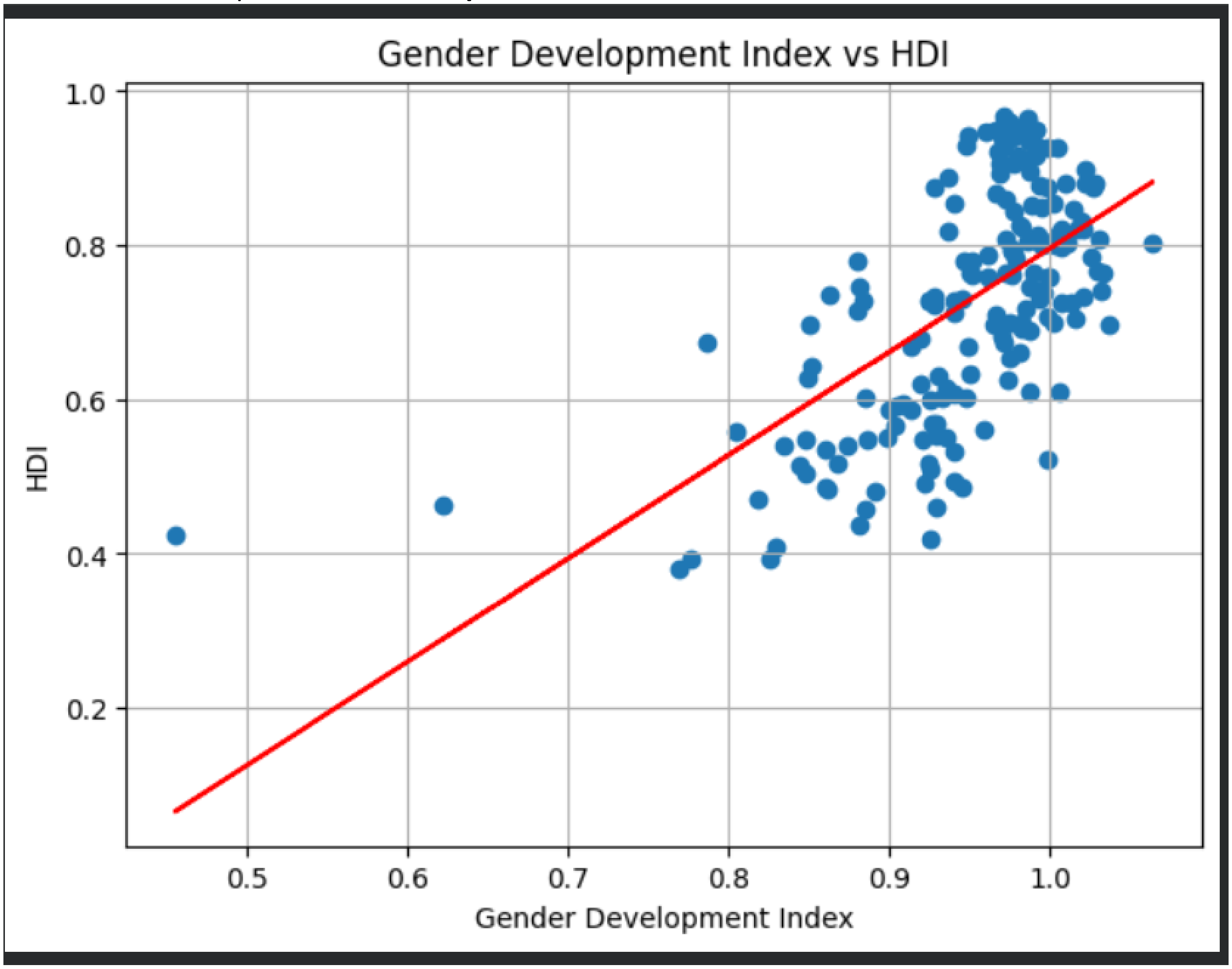
- Maldives, Sri Lanka, and Bhutan achieved the highest composite scores
- There is a significant positive association between life expectancy and HDI.
- There is a somewhat favorable correlation between the Gender Development Index and HDI.
- There are significant positive GNI–HDI gaps in nations like Sri Lanka and the Maldives.

Visualizations and Tables:

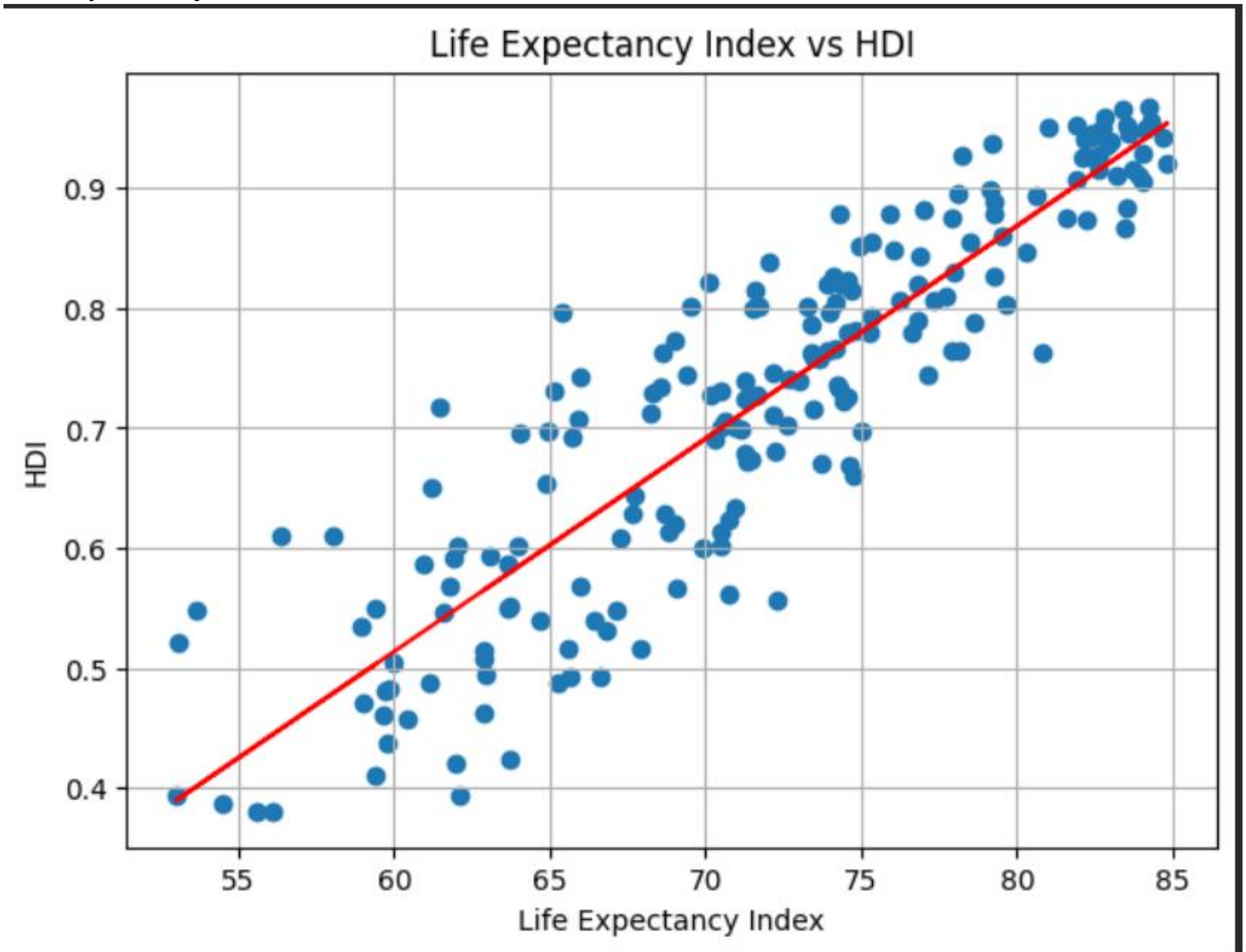
Bar chart of top South Asian countries by composite score:



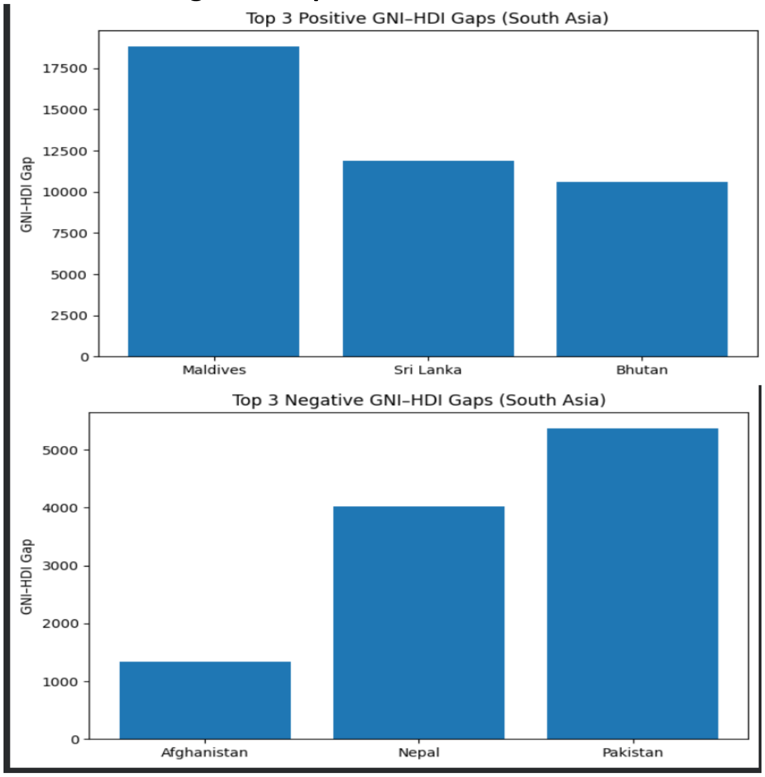
Metric Relationships **Gender Development vs HDi:**



Life Expectancy vs HDI:



Positive & Negative Gaps:



Interpretation and Discussion:

In South Asia, life expectancy—a measure of health outcomes—is a key factor in determining HDI. Higher development results are not guaranteed by income alone, underscoring the significance of balanced advancement across social metrics.

25%

Overall Similarity

1	University of Wolverhampton on 2026-01-08 SUBMITTED WORKS	4%
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2.4 Problem 3

Approach and methodology:

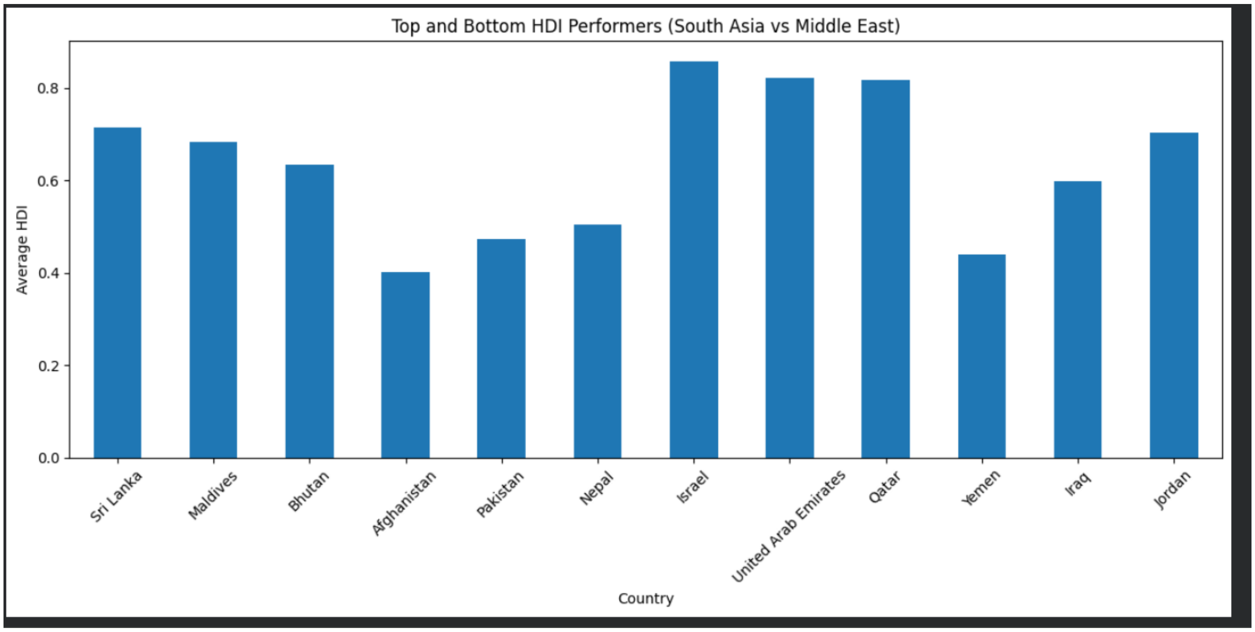
South Asia and the Middle East were compared using HDI data from 2020 to 2022. Geographical discrepancies were evaluated using descriptive statistics, top and bottom performance rankings, correlation analysis, and outlier detection.

Key Results:

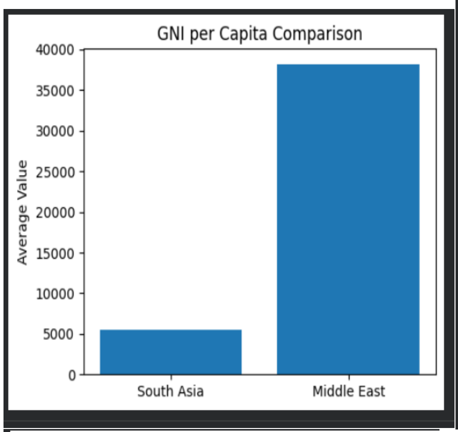
- The average HDI is greater in the Middle East than in South Asia.
- The Middle East has more economic heterogeneity because of extreme income values; South Asia has more relative diversity in HDI; and both regions have strong positive connections between HDI and life expectancy.

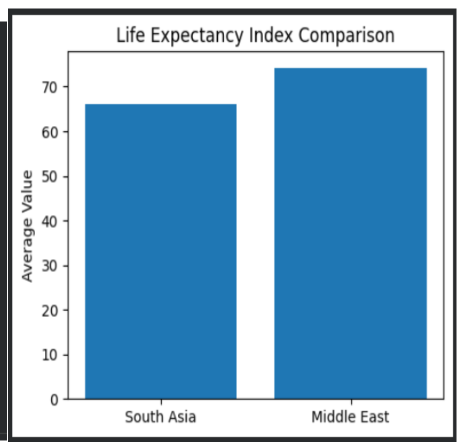
Visualizations and Tables:

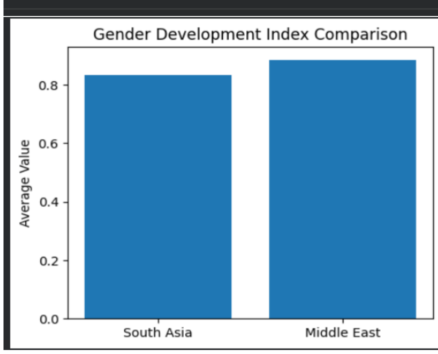
Bar charts comparing top and bottom HDI performers:



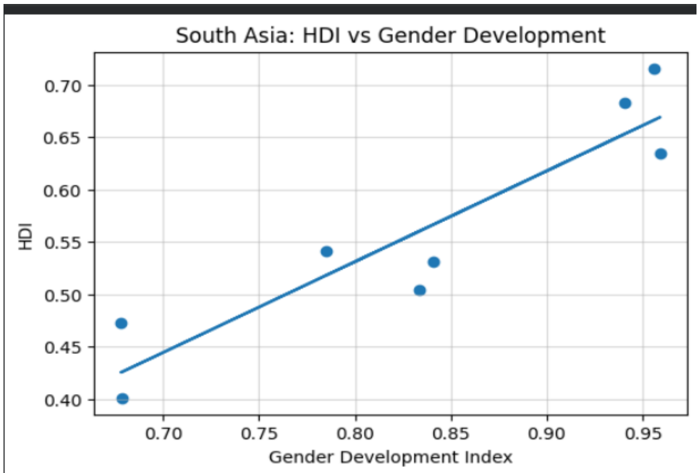
Metric Comparisons:

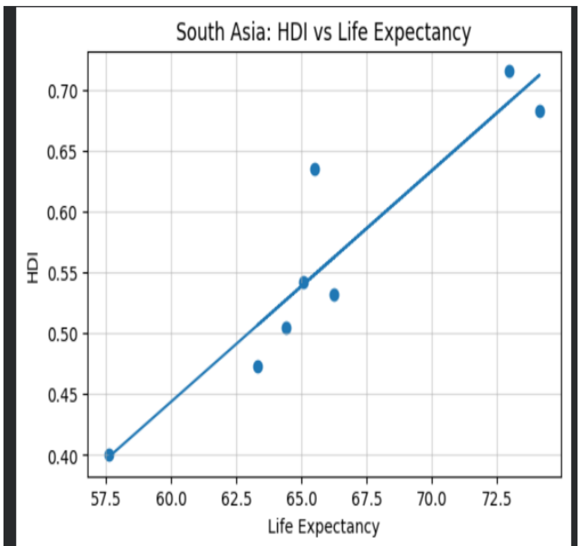


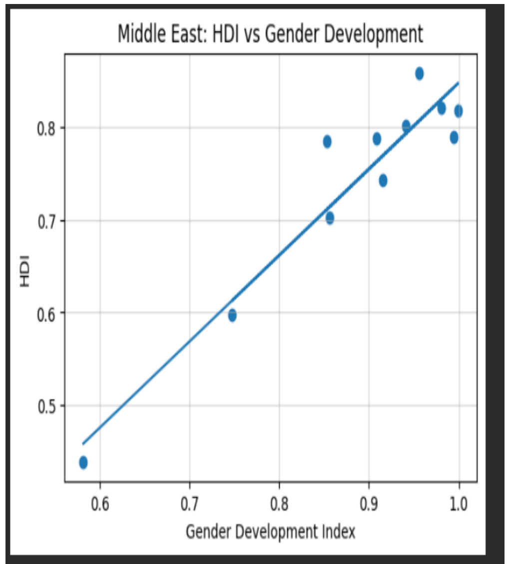


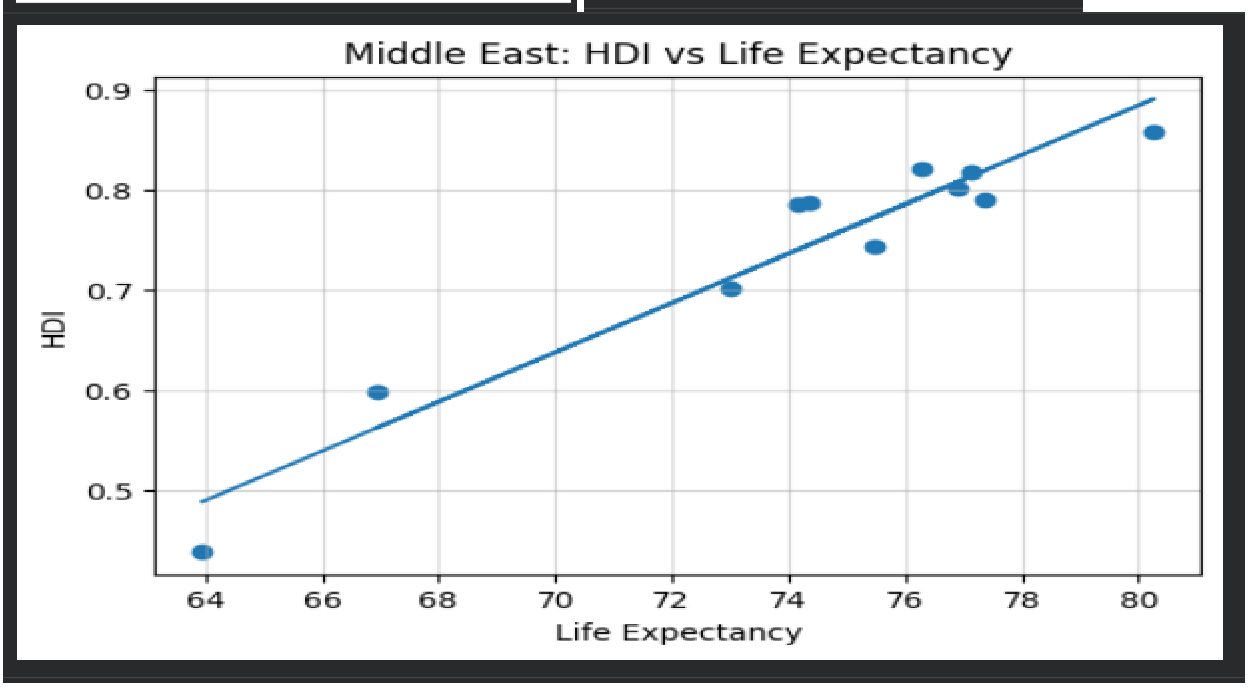


correlations of HDI with:

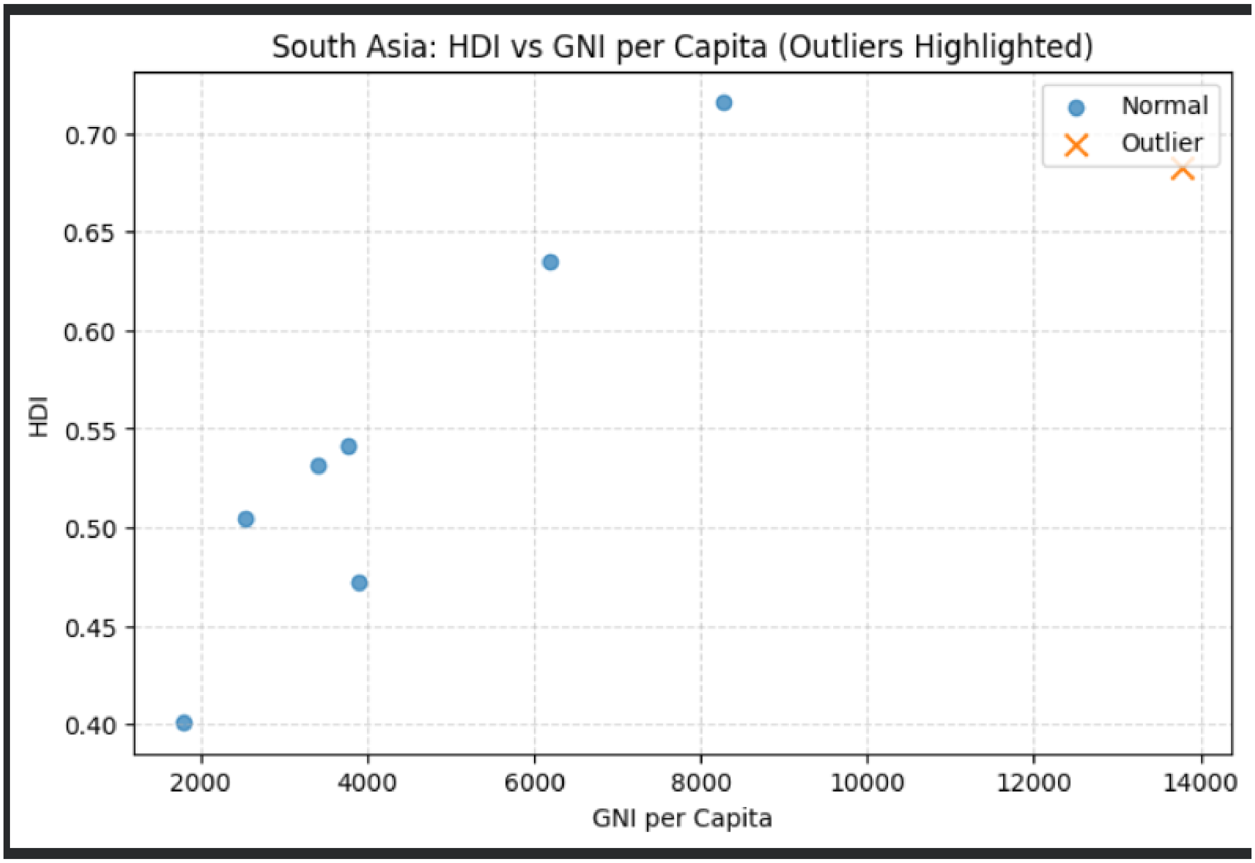

South Asia: HDI vs Gender Development
HDI
0.70
0.65
0.60
0.55
0.50
0.45
0.40
0.70
0.75
0.80
0.85
0.90
0.95
Gender Development Index


South Asia: HDI vs Life Expectancy
HDI
0.70
0.65
0.60
0.55
0.50
0.45
0.40
57.5
60.0
62.5
65.0
67.5
70.0
72.5
Life Expectancy

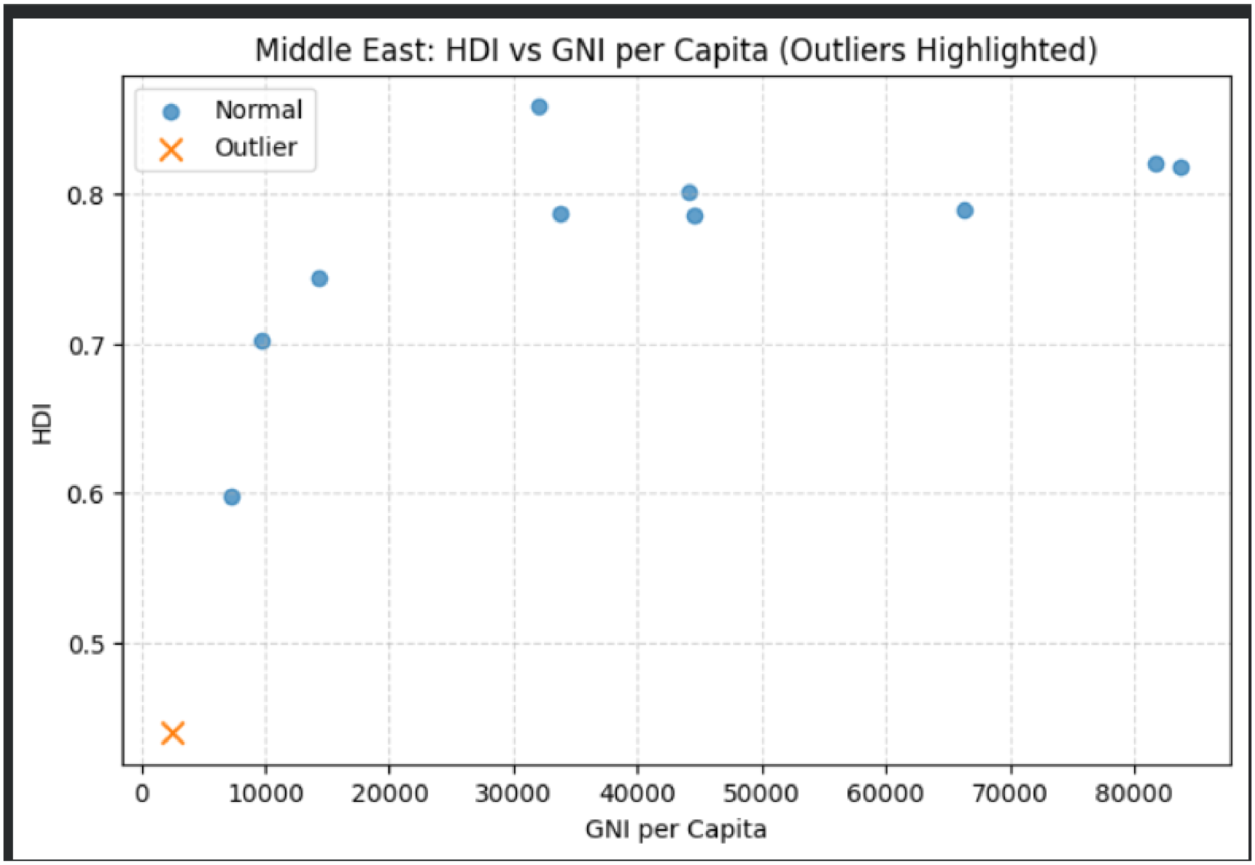

Middle East: HDI vs Gender Development
HDI
0.8
0.7
0.6
0.5
0.6
0.7
0.8
0.9
1.0
Gender Development Index


Middle East: HDI vs Life Expectancy
HDI
0.9
0.8
0.7
0.6
0.5
64
66
68
70
72
74
76
78
80
Life Expectancy

Outlier:



South Asia: HDI vs GNI per Capita (Outliers Highlighted)
Normal
Outlier
HDI
0.70
0.65
0.60
0.55
0.50
0.45
0.40
2000
4000
6000
8000
10000
12000
14000
GNI per Capita



Interpretation and Discussion:

While Middle Eastern countries generally achieve higher HDI levels, development outcomes vary widely due to economic diversity. South Asia shows more uniform but lower development levels, emphasizing the need for targeted improvements in health and education.

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3. **Conclusion**

Summary of Findings

- Global HDI shows gradual improvement from 2020 to 2022
- Significant disparities persist across regions and income groups
- Health indicators have the strongest relationship with HDI
- Regional comparisons reveal distinct development challenges

Limitations

- Analysis limited to three years of data
- Some indicators, such as education subcomponents, were not deeply explored
- HDI does not capture within-country inequalities

Recommendations

- Developing regions should prioritize health and education investments
- Policymakers should use composite indicators alongside HDI
- Future studies could extend the analysis to more recent years

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References

- The United Nations Development Programme (UNDP). Reports on Human Development
- Course lecture notes & assignment dataset.

- Link: https://github.com/np03cs4a240277-alt/Assesment.git

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